

Write your name here	
Surname	Other names
Pearson Edexcel International Advanced Level	Centre Number
Candidate Number 	
Physics Advanced Subsidiary Unit 2: Physics at Work	
Thursday 4 June 2015 – Afternoon Time: 1 hour 30 minutes	Paper Reference WPH02/01
You must have: Ruler, protractor	Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The total mark for this paper is 80.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*
- Questions labelled with an **asterisk** (*) are ones where the quality of your written communication will be assessed
– *you should take particular care with your spelling, punctuation and grammar, as well as the clarity of expression, on these questions.*
- The list of data, formulae and relationships is printed at the end of this booklet.
- Candidates may use a scientific calculator.

Advice

- Read each question carefully before you start to answer it.
- Keep an eye on the time.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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PEARSON

SECTION A

Answer ALL questions.

For questions 1–10, in Section A, select one answer from A to D and put a cross in the box ☐. If you change your mind, put a line through the box ☒ and then mark your new answer with a cross ☐.

1 Which of the following is an SI base unit?

- ☐ A ampere
☐ B charge
☐ C coulomb
☐ D current

(Total for Question 1 = 1 mark)

2 A cell with emf $\mathcal{E}$ and internal resistance r has a potential difference V across its terminals when the current is I .

The internal resistance r is given by

- ☐ A $\frac{\mathcal{E}}{I}$
☐ B $\frac{V}{I}$
☐ C $\frac{\mathcal{E} + V}{I}$
☐ D $\frac{\mathcal{E} - V}{I}$

(Total for Question 2 = 1 mark)

3 A rechargeable cell is labelled 1500 mA h.

If the current is 1500 mA for 1 hour, the charge transferred is

- ☐ A 1.5 C
☐ B 90 C
☐ C 1500 C
☐ D 5400 C

(Total for Question 3 = 1 mark)

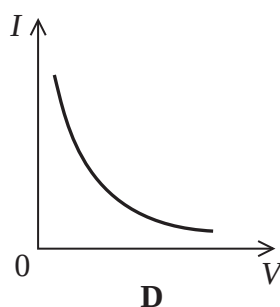
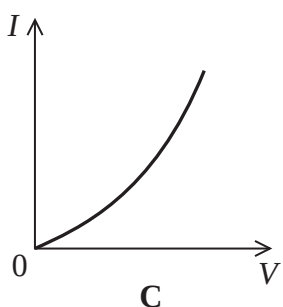
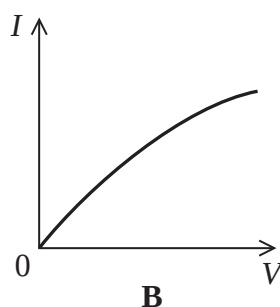
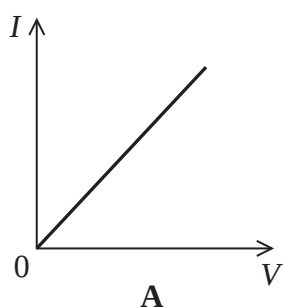


4 The level of detail in an ultrasound scan can be increased by using a

- ☐ A higher frequency.
- ☐ B higher wave speed.
- ☐ C longer pulse duration.
- ☐ D longer wavelength.

(Total for Question 4 = 1 mark)

5 Which graph correctly shows how the current I varies with the potential difference V for a negative temperature coefficient thermistor?



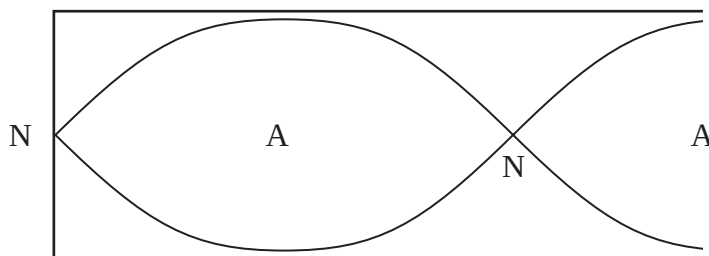
- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 5 = 1 mark)



Questions 6 and 7 refer to the diagram below.

The diagram represents a stationary sound wave in a pipe closed at one end.



Nodes are labelled N and antinodes are labelled A.

6 Which of the following statements is true?

- ☐ **A** The amplitude at a node is always a maximum.
- ☐ **B** The antinodes are positions of maximum amplitude.
- ☐ **C** The displacement at adjacent nodes is in opposite directions.
- ☐ **D** The displacement is perpendicular to the direction of wave travel.

(Total for Question 6 = 1 mark)

7 The length of the pipe is 0.75 m.

What is the wavelength of the stationary wave?

- ☐ **A** 0.25 m
- ☐ **B** 0.50 m
- ☐ **C** 1.00 m
- ☐ **D** 1.50 m

(Total for Question 7 = 1 mark)



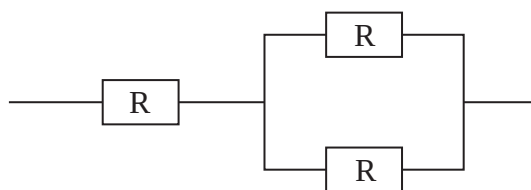
- 8 When a charge of 2.0 C passes through a light bulb, 5.0 J of energy is transferred.

What is the potential difference across the bulb?

- ☐ A 0.4 V
- ☐ B 2.5 V
- ☐ C 3.0 V
- ☐ D 10 V

(Total for Question 8 = 1 mark)

- 9 The diagram shows a combination of three resistors each of resistance R .



The total resistance of the combination can be found using

- ☐ A $R + \frac{1}{R} + \frac{1}{R}$
- ☐ B $R + \frac{1}{\left(\frac{1}{R} + \frac{1}{R}\right)}$
- ☐ C $\frac{1}{\left(\frac{1}{R} + \frac{1}{(R + R)}\right)}$
- ☐ D $\frac{1}{R} + \frac{1}{\left(\frac{1}{R} + \frac{1}{R}\right)}$

(Total for Question 9 = 1 mark)



10 An ambulance is moving towards an observer with its siren sounding.

Which row of the table correctly describes the apparent changes in wave properties of the sound from the siren caused by the Doppler effect?

	Wave speed	Frequency	Wavelength
☐ A	no change	increases	decreases
☐ B	decreases	decreases	increases
☐ C	no change	decreases	increases
☐ D	increases	increases	decreases

(Total for Question 10 = 1 mark)

TOTAL FOR SECTION A = 10 MARKS



SECTION B**Answer ALL questions in the spaces provided.****11** The list of data, formulae and relationships states

$$\text{Current} \qquad I = nqvA$$

Show that the units on each side of the equation are consistent.

(3)

(Total for Question 11 = 3 marks)



- 13** Surveyors sometimes use laser rangefinders to measure the distance to objects such as buildings and trees.

A reflector is placed on the object. The rangefinder emits pulses of light and detects them when they return after being reflected.

- (a) State why the laser light is emitted in pulses.

(1)

- (b) The rangefinder measures distances between 50 cm and 1 km.

Calculate the longest pulse duration that would allow this range of measurements.

(3)

Pulse duration =

- (c) Distances inside buildings, such as the length of a room, are often measured using ultrasound.

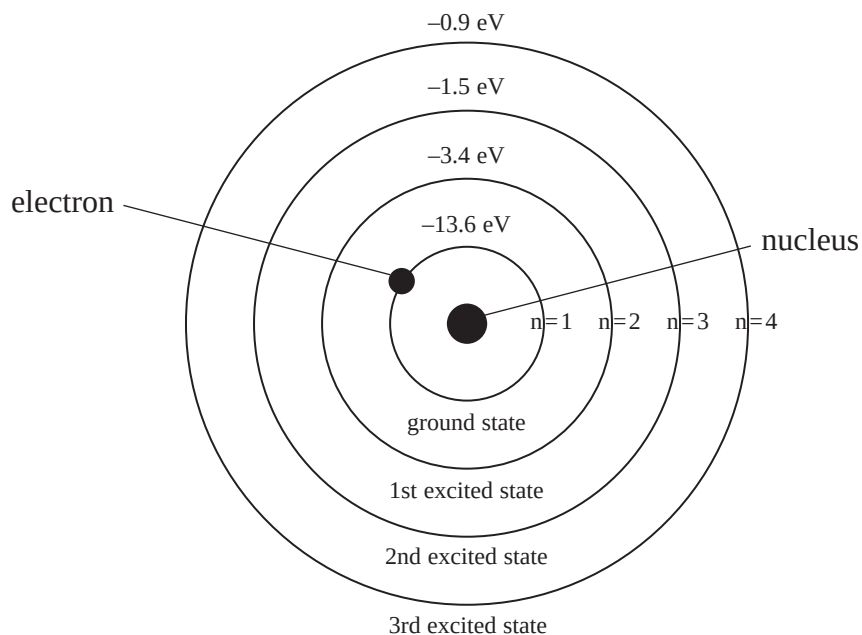
Suggest a reason why a laser rangefinder would be more suitable than one using ultrasound for measuring the distance to a tree 1 km away.

(1)

(Total for Question 13 = 5 marks)



15 In 1913 Niels Bohr proposed the model of the hydrogen atom represented in the diagram.



In this model an electron can orbit the nucleus at different energy levels, some of which are shown in the diagram.

(a) State what is meant by excited with reference to this model.

(1)

(b) Calculate the highest frequency of radiation that could be emitted by electrons involved in transitions between energy levels shown in the diagram.

(4)

Frequency =

(Total for Question 15 = 5 marks)



16 (a) State what is meant by the principle of superposition of waves.

(2)

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(b) A teacher demonstrating superposition set up two speakers in a laboratory. The students stood in different positions throughout the laboratory. The teacher played a single note through one of the speakers.

The teacher then played the note through both speakers and asked the students to describe their observations. Students in some positions said the sound got louder and students in other positions said the sound got quieter.

The students noted positions of louder sound L and quieter sound Q. Their results are shown in the diagram.

Q L Q L Q L Q

Q L Q L Q L Q

Q L₁ Q₁ L Q L Q

X

Y

Not to scale

speakers



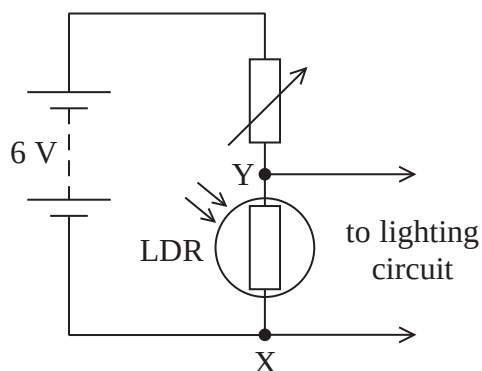


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P 4 4 9 2 5 A 0 1 5 2 8

- 18** A student designed a circuit to switch on a light when it gets dark. The circuit contained a light dependent resistor (LDR) in series with a variable resistor to control the light level at which the lighting circuit is switched on.



The resistance of the LDR decreases as the radiation flux of the incident light increases.

The student placed a lamp at different distances from the LDR. For each distance the radiation flux incident on the LDR was measured using a light meter and the resistance of the LDR was measured.

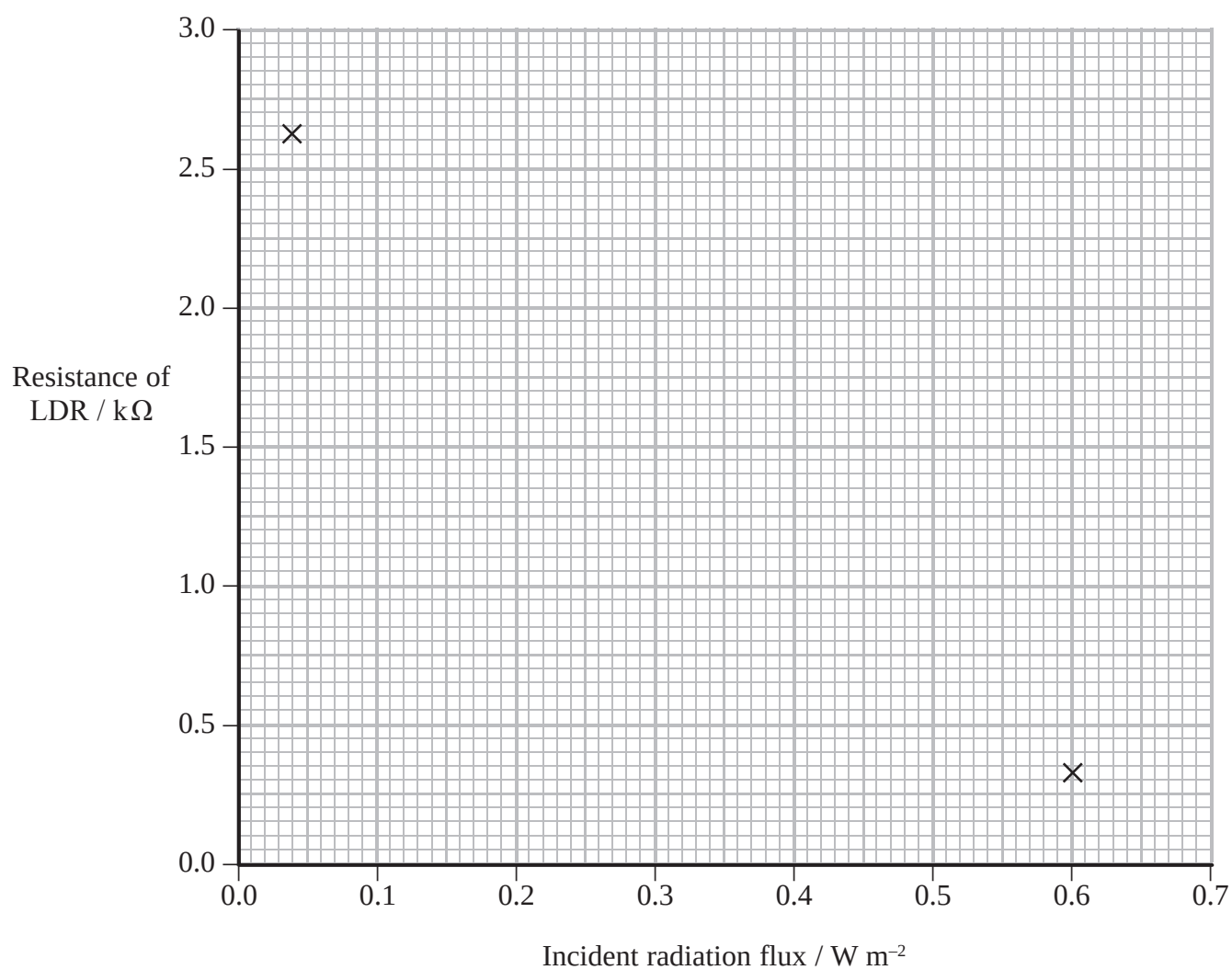
The results are shown in the table.

Incident radiation flux / W m^{-2}	Resistance of LDR / $\text{k}\Omega$
0.04	2.62
0.09	1.37
0.11	1.08
0.17	0.63
0.35	0.44
0.60	0.32



(a) Use the results in the table to complete the graph.

(3)



(b) The lighting circuit will switch on when the potential difference across XY is 0.60 V.

Determine the required resistance R of the variable resistor so that the lighting circuit will switch on when the incident radiation flux is 0.25 W m^{-2} .

(3)

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Resistance =

(c) Apart from eliminating human error, suggest how using a resistance sensor and a radiation flux sensor connected to a data logger could have improved the quality of the graph.

(2)

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(Total for Question 18 = 8 marks)

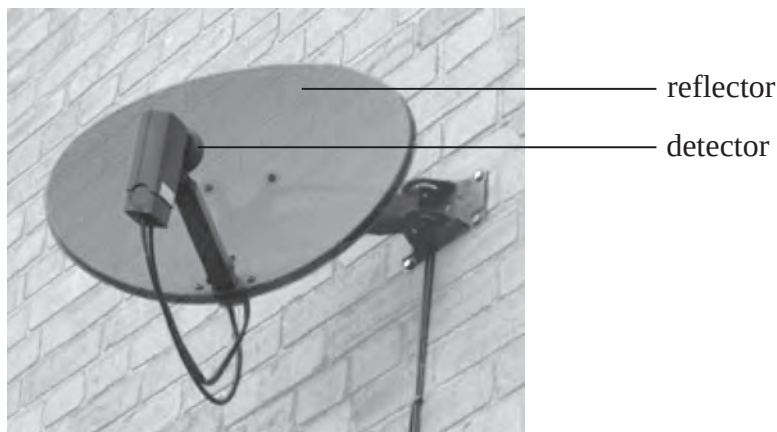


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P 4 4 9 2 5 A 0 1 9 2 8

19 The photograph shows a satellite television dish.



Electromagnetic radiation from a communications satellite is reflected from the reflector to the detector.

(a) The radiation used has a frequency of 12.6 GHz.

(i) Show that the wavelength of the radiation is about 2 cm.

(2)

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(ii) State the region of the electromagnetic spectrum to which this radiation belongs.

(1)

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(b) The radiation incident on the reflector has a radiation flux of $4.8 \times 10^{-13} \text{ W m}^{-2}$.

Calculate the power of the incident radiation.

area of the reflector = 0.27 m^2

(2)

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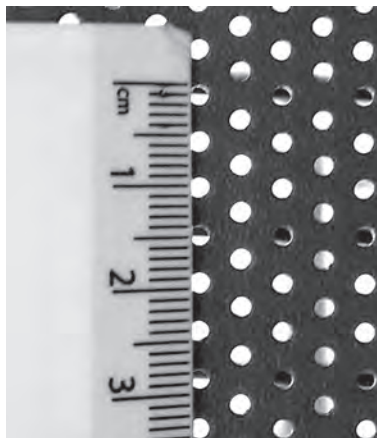
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Power =



(c) The reflector contains many small holes.

(i) Use the photograph to estimate the diameter of the holes.



(1)

(ii) It is important that the radiation is reflected to the detector with the maximum possible power.

Use the idea of diffraction effects to explain why the radiation is reflected as if from solid metal.

(2)

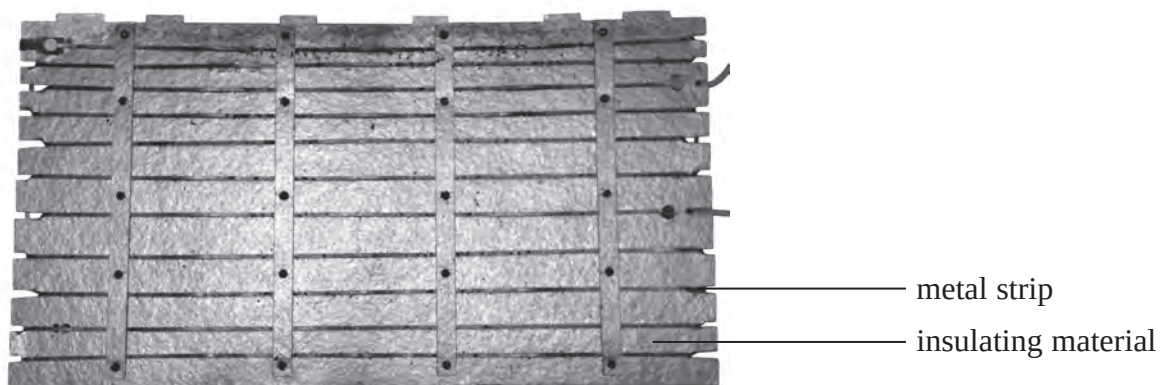
(iii) Suggest a reason for having holes in the reflector, rather than using solid metal.

(1)

(Total for Question 19 = 9 marks)

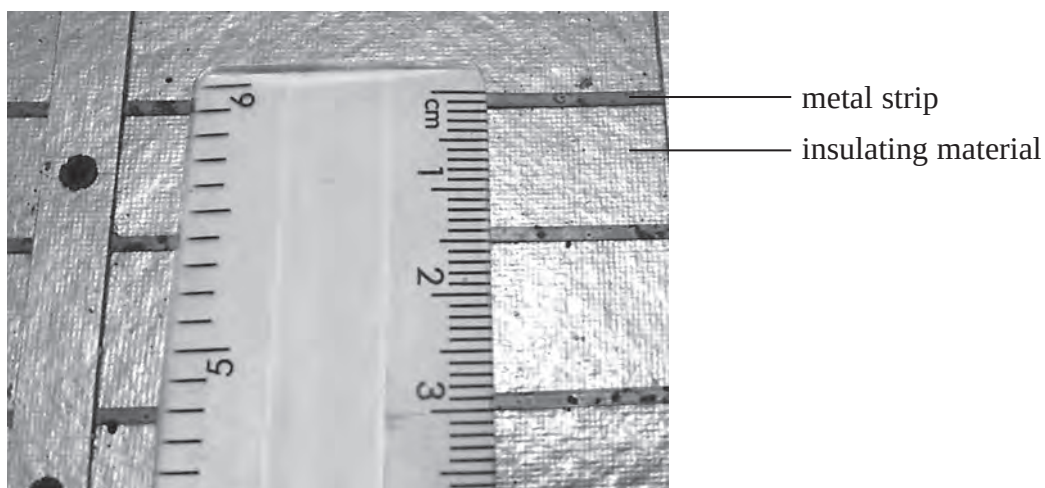


- 20 The photograph shows a heating section from a toaster. This consists of a long, thin, metal strip mounted on an insulating material.



The metal strip carries a large current producing a high temperature. As a result, infrared radiation is emitted which toasts bread.

- (a) A student is asked to identify the metal used for the strip by calculating its resistivity. The student takes measurements of the resistance and dimensions of the metal strip.
- (i) The photograph shows the student measuring the width of the metal strip.



Comment on the use of this method to measure the width of the metal strip.

(2)

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(ii) The student obtained the following measurements for a sample of the metal strip.

resistance = 1.8Ω

length = 48.5 cm

width = 1.5 mm

thickness = 0.24 mm

Calculate the resistivity of the metal in $\Omega \text{ m}$.

(2)

Resistivity = $\Omega \text{ m}$

(b) The following information is shown on the toaster.

1400 W, 230 V

Calculate the total resistance of the metal strips in the toaster.

(2)

Total resistance =

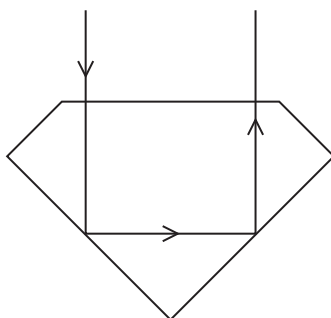
(c) Explain why the resistance of the metal strips in the toaster increases as the temperature increases.

(3)

(Total for Question 20 = 9 marks)



- 21 Cut gemstones used in jewellery ‘sparkle’ because a large proportion of the incident light undergoes total internal reflection. An example of total internal reflection in a cut gemstone is shown in the diagram.



- (a) Diamond is a popular gemstone because it has a very high refractive index.

Show that the refractive index of diamond is about 2.4.

speed of light in diamond = $1.24 \times 10^8 \text{ m s}^{-1}$

(2)

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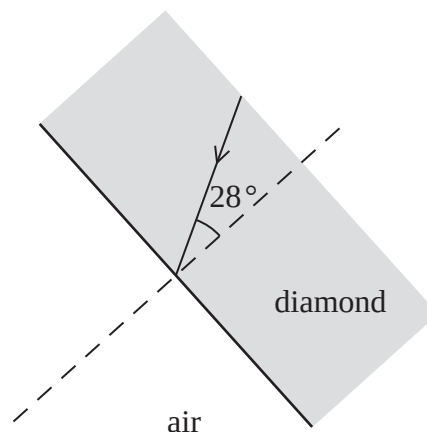
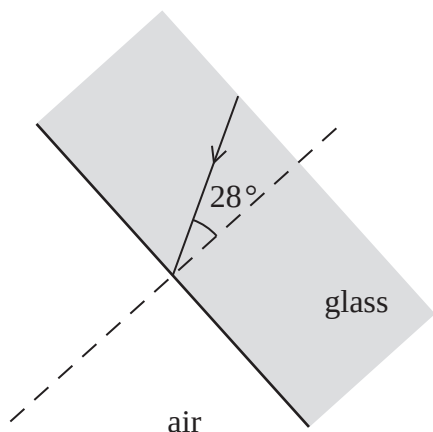
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(b) Imitation gemstones can be made from glass.

The diagrams show incident light at a boundary of glass with air and at a boundary of diamond with air. The angle of incidence is 28° in each case.



Use appropriate calculations to determine what happens to the light in each case and complete the diagrams to show this, labelling the relevant angles.

refractive index of glass = 1.50

(6)

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(c) Suggest why diamonds sparkle more than imitations made from glass.

(2)

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(Total for Question 21 = 10 marks)

TOTAL FOR SECTION B = 70 MARKS

TOTAL FOR PAPER = 80 MARKS



List of data, formulae and relationships

Acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$	(close to Earth's surface)
Electron charge	$e = -1.60 \times 10^{-19} \text{ C}$	
Electron mass	$m_e = 9.11 \times 10^{-31} \text{ kg}$	
Electronvolt	$1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$	
Gravitational field strength	$g = 9.81 \text{ N kg}^{-1}$	(close to Earth's surface)
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$	
Speed of light in a vacuum	$c = 3.00 \times 10^8 \text{ m s}^{-1}$	

Unit 1

Mechanics

Kinematic equations of motion	$v = u + at$ $s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
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Forces	$\Sigma F = ma$ $g = F/m$ $W = mg$
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Work and energy	$\Delta W = F\Delta s$ $E_k = \frac{1}{2}mv^2$ $\Delta E_{\text{grav}} = mg\Delta h$
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Materials

Stokes' law	$F = 6\pi\eta rv$
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Hooke's law	$F = k\Delta x$
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Density	$\rho = m/V$
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Pressure	$p = F/A$
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Young modulus	$E = \sigma/\epsilon$ where Stress $\sigma = F/A$ Strain $\epsilon = \Delta x/x$
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Elastic strain energy	$E_{\text{el}} = \frac{1}{2}F\Delta x$
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Unit 2

Waves

Wave speed

$$v = f\lambda$$

Refractive index

$${}_1\mu_2 = \sin i / \sin r = v_1 / v_2$$

Electricity

Potential difference

$$V = W/Q$$

Resistance

$$R = V/I$$

Electrical power, energy and efficiency

$$P = VI$$

$$P = I^2R$$

$$P = V^2/R$$

$$W = VIt$$

$$\% \text{ efficiency} = \frac{\text{useful energy output}}{\text{total energy input}} \times 100$$

$$\% \text{ efficiency} = \frac{\text{useful power output}}{\text{total power input}} \times 100$$

Resistivity

$$R = \rho l/A$$

Current

$$I = \Delta Q / \Delta t$$

$$I = nqvA$$

Resistors in series

$$R = R_1 + R_2 + R_3$$

Resistors in parallel

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Quantum physics

Photon model

$$E = hf$$

Einstein's photoelectric equation

$$hf = \phi + \frac{1}{2}mv_{\max}^2$$

